

Unit 2 Homework — Exponents & Scientific Notation

Complete independently · Exponent Rules · Scientific Notation

A. Exponent Rules — Product & Quotient

Problem 1

Simplify: $x^5 \cdot x^6$

$$x^{11}$$

Problem 2

Simplify: $b^9 \div b^4$

$$b^5$$

Problem 3

Simplify: $(4x^2)(2x^5)$

$$8x^7$$

Problem 4

Simplify: $24m^8 \div 6m^3$

$$4m^5$$

Problem 5

Simplify: $c^3 \cdot c^4 \cdot c^2$

$$c^9$$

Problem 6

Simplify: $(7y^4)(2y)$

$$14y^5$$

B. Power, Zero & Negative Exponents

Problem 1

Simplify: $(y^5)^2$

$$y^{10}$$

Problem 2

Simplify: $(3x^2)^2$

$$9x^4$$

Problem 3

Simplify: 12^0

1

Problem 4

Simplify: $(8m^3n^2)^0$

1

Problem 5

Simplify: 2^{-3} (write as a fraction)

$1/8$

Problem 6

Simplify: y^{-2} (write as a fraction)

$1/y^2$

Problem 7

Evaluate -2^4 (no parentheses).

-16

C. Scientific Notation — Conversions

Problem 1

Write 81,000 in scientific notation.

8.1×10^4

Problem 2

Write 0.00056 in scientific notation.

5.6×10^{-4}

Problem 3

Write 9.4×10^5 in standard form.

940,000

Problem 4

Write 6.02×10^{-3} in standard form.

0.00602

Problem 5

Write 12,300,000 in scientific notation.

$$\underline{1.23 \times 10^7}$$

D. Scientific Notation — Operations**Problem 1**

Multiply: $(4 \times 10^2) \times (2 \times 10^6)$

$$\underline{8 \times 10^8}$$

Problem 2

Divide: $(9 \times 10^8) \div (3 \times 10^2)$

$$\underline{3 \times 10^6}$$

Problem 3

Multiply: $(5 \times 10^3) \times (6 \times 10^4)$ — adjust if needed!

$$\underline{3 \times 10^8}$$

E. Word Problem**Problem 1**

Light travels 1.86×10^5 miles per second. How far does it travel in 10 seconds? (scientific notation)

$$\underline{1.86 \times 10^6}$$